

HOUSE PRICE PREDICTION USING MACHINE LEARNING

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Project Objective

The objective of this project was to develop a machine learning model capable of predicting house prices based on various housing characteristics such as area, number of bedrooms, bathrooms, stories, parking facilities, and additional amenities. The project aimed to identify the key factors influencing house prices and evaluate the effectiveness of different regression models for accurate price prediction.

Dataset Description

The dataset used in this project was the Housing Prices Dataset obtained from Kaggle. It contains information about residential properties, including area, bedrooms, bathrooms, stories, parking availability, furnishing status, and other features that may influence house prices. The target variable for prediction was the house price.

Data Preprocessing

The dataset was explored and cleaned before model development. Missing values and duplicate records were checked and handled appropriately. Categorical variables were converted into numerical format using one-hot encoding to make them suitable for machine

learning algorithms. The cleaned dataset was then prepared for model training and evaluation.

Exploratory Data Analysis

Several visualizations were created to understand the data and identify important patterns. These included house price distribution, correlation heatmaps, price versus area analysis, feature importance visualization, furnishing status analysis, and actual versus predicted price comparisons. The analysis helped identify relationships between housing features and property prices.

Models Used

Two regression models were developed and evaluated:

- 1. Linear Regression**
- 2. Random Forest Regressor**

The dataset was split into training and testing sets using an 80:20 ratio. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Model Performance

Linear Regression

- **MAE: 970,043**
- **RMSE: 1,324,507**
- **R^2 Score: 0.653**

Random Forest Regressor

- **MAE: 1,021,546**
- **RMSE: 1,400,566**
- **R^2 Score: 0.612**

The Linear Regression model achieved better performance across all evaluation metrics and was selected as the best-performing model for this project.

Key Findings

Feature importance analysis revealed that area was the most influential factor affecting house prices, contributing nearly 47% of the model's predictive power. Bathrooms, air conditioning, parking availability, stories, and bedrooms also played important roles in determining property value. The analysis showed that both structural characteristics and lifestyle-related amenities significantly influence housing prices.

Recommendation

Based on the findings, real estate businesses should focus on property size, number of bathrooms, parking facilities, and premium amenities when pricing and marketing residential properties. Highlighting these features can improve property valuation accuracy and attract potential buyers more effectively.

Conclusion

This project successfully demonstrated the application of machine learning techniques for house price prediction. Through data preprocessing, exploratory analysis, visualization, and model evaluation, meaningful insights were obtained regarding the factors influencing house prices. The Linear Regression model achieved the best performance with an R^2 score of 0.653, showing that housing characteristics can be effectively used to estimate property values and support data-driven decision-making in the real estate industry.